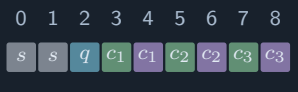
$$\widehat{H} = H[P,:], \quad U = \mathrm{GELU}(\widehat{H}W_1^{\mathsf{T}} + b_1), \quad s = \mathrm{squeeze}(UW_2^{\mathsf{T}} + b_2)$$

1. One causal candidate list



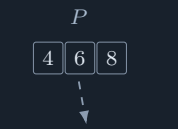
x
T integer token IDs

2. Qwen3-0.6B-Base

28 causal decoder blocks
RMSNorm → GQA
RMSNorm → gated FFN
residuals;
LoRA $r = 16$

H
$T \times d$

3. Candidate-end gather

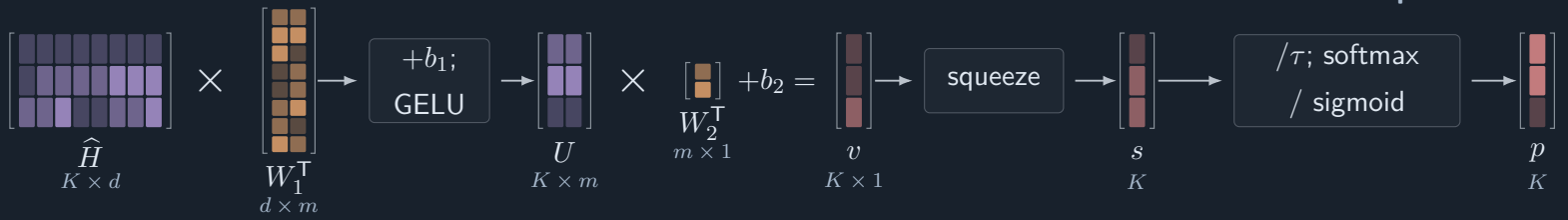


$\widehat{H}$
$K \times d$

Candidate visibility

$M^{\mathrm{cand}\to\mathrm{cand}}$
$K \times K$
row = query; column = key
later candidates see earlier ones

4. Shared scalar MLP: $1024 \to 256 \to 1$



$\widehat{H}$ $K \times d$ · W_1^{T} $d \times m$ · U $K \times m$ · W_2^{T} $m \times 1$ · v $K \times 1$ · s K

5. Normalize over the valid answer space

/τ; softmax / sigmoid

p
K

Axes One question shown; batch B omitted. T: tokens; K: valid candidates; $d = 1024$, $m = 256$. Tiles illustrate axis structure, not the literal dimensions.

Objects x and P are integer IDs/positions; $H, \widehat{H}, U$ are continuous states; W_1, W_2 are learned weights; s are raw scores and p probabilities. Biases appear in the formula.

Mechanism Choice example, `candidate_scalar/list`: actual candidate-end tokens, no inserted markers. Noul instead selects the final Answer cue ($K = 1$). Choice/score use masked softmax; noul/multi-label use sigmoid. No tree cache; LM head bypassed.